

MITRE ATLAS v5.1 FRAMEWORK

Adversarial AI/ML Threat Landscape & Agentic Security Matrix

DOCUMENT SCOPE & TAXONOMY OVERVIEW

This reference manual outlines the complete structural matrix of the **MITRE ATLAS v5.1 Adversarial AI/ML threat framework** (released November 2025). Modeled after the industry-standard MITRE ATT&CK framework, ATLAS provides security architects, red teams, and threat hunters with a comprehensive, community-validated database of adversary tactics, techniques, and real-world incidents targeting machine learning systems.

The v5.1.0 update marks a critical transition in AI security from model-centric input attacks to **autonomous AI agent runtime security**. Developed in collaboration with Zenity Labs, the framework introduces 14 formal agent-focused techniques covering memory manipulation, context poisoning, tool-chain abuse, and covert C2 infrastructure. The target scope spans training pipelines, model weights, inference APIs, vector databases (RAG), and connected agent environments.

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TACTICS	TECHNIQUES	SUB-TECHNIQUES	CASE STUDIES / MITIGATIONS

ATT&CK Architectural Alignment & AI-Specific Tactics

While ATLAS inherits and adapts 14 tactics from traditional enterprise IT security to AI workloads, it maintains **two completely unique tactics** not found in ATT&CK: **ML Model Access (AML.TA0004)** and **ML Attack Staging (AML.TA0012)**. These represent the specific access requirements and the training/validation-phase preparations (such as training data poisoning or backdoor insertion) necessary to compromise non-deterministic machine learning architectures.

THE 16 ADVERSARIAL TACTICS OF MITRE ATLAS

Tactics in MITRE ATLAS represent the high-level adversary objectives across an attack lifecycle. In the November 2025 release (v5.1.0), the additions of **Lateral Movement (AML.TA0015)** and **Command and Control (AML.TA0014)** expanded the matrix to 16 tactics, mapping the entire AI compromise pathway from initial reconnaissance to permanent impact.

Tactic ID	ATLAS Tactic Name	Strategic Objective / Description	Representative Techniques
AML.TA0002	Reconnaissance	Active or passive gathering of target AI system capabilities, model architecture, or ontology.	Discover ML Artifacts, Discover ML Model Ontology, Active Scanning
AML.TA0003	Resource Development	Preparing tools, surrogate models, or datasets to support adversarial campaign execution.	Acquire Public ML Artifacts, Develop Adversarial ML Attack Capabilities
AML.TA0004	Initial Access	Gaining an initial entry point into the target AI model runtime or operational network.	Prompt Injection (AML.T0051), ML Supply Chain Compromise
AML.TA0000	ML Model Access	Gaining query-level or direct access to the model weights or inference APIs (Unique to AI).	Inference API Access, ML Artifacts Access
AML.TA0005	Execution	Running unauthorized commands, prompt-driven tasks, or scripting inside the agent runtime.	LLM Plugin Compromise, AI Agent Tool Invocation (AML.T0053)
AML.TA0006	Persistence	Maintaining long-term access and control across sessions through AI state manipulation.	Modify AI Agent Configuration (AML.T0081), AI Agent Context Poisoning (AML.T0080)
AML.TA0012	Privilege Escalation	Elevating capabilities or escaping execution boundaries to perform unauthorized actions.	Exploit through ML System, Escaping to Host, Intent Goal Hijacking
AML.TA0007	Defense Evasion	Avoiding detection by obfuscating inputs, bypassing guardrails, or manipulating safety flags.	LLM Jailbreak (AML.T0054), Prompt Obfuscation (AML.T0068), Adversarial Perturbation
AML.TA0013	Credential Access	Harvesting credentials or API keys stored in configuration files or tool boundaries.	Credentials from AI Agent Configuration, Tool Credential Harvesting (AML.T0098)
AML.TA0008	Discovery	Probing the environment to discover agent config, tool registries, or active processes.	Discover AI Agent Configuration (AML.T0084), Process Discovery
AML.TA0015	Lateral Movement	Moving from a compromised agent or system to other connected enterprise systems (Added Nov 2025).	Move through Agent tool chains, MCP Server Pivot
AML.TA0009	Collection	Gathering sensitive data, prompt inputs, or vector database embeddings from the AI ecosystem.	RAG Database Retrieval, Data from AI Services (AML.T0085)

Tactic ID	ATLAS Tactic Name	Strategic Objective / Description	Representative Techniques
AML.TA0001	ML Attack Staging	Preparing adversarial inputs or training data poisoning (Unique to AI).	Poison Training Data (AML.T0020), Backdoor ML Model
AML.TA0014	Command & Control	Repurposing AI orchestration or service APIs as a covert control channel (Added Nov 2025).	Reverse Shell, LLM API Covert C2 (SesameOp AML.CS0042)
AML.TA0010	Exfiltration	Extracting training datasets, proprietary model weights, or system credentials.	Exfiltration via ML Inference API, Exfiltration via Tool Invocation (AML.T0086)
AML.TA0011	Impact	Disrupting, manipulating, or destroying AI systems, data, or operational capabilities.	Denial of ML Service, Data Destruction via Tool Invocation (AML.T0101)

THE AGENTIC AI & SYSTEM-STATE THREAT MATRIX

Autonomous AI agents that execute shell tools, access corporate filesystems, and read/write their own configuration and memory create a massive new attack surface. Unlike traditional static apps, agents read their persistent state as the direct context for future decisions. In collaboration with Zenity Labs, MITRE ATLAS formalized 14 new techniques to catalog how attackers can compromise these autonomous workflows:

Tech ID	ATLAS Technique Name	Strategic Tactic	Threat Vector & Technical Description
AML.T0080	AI Agent Context Poisoning	Persistence	Injecting adversarial instructions into memory/thread state. Sub-techniques: Memory (AML.T0080.000) and Thread (AML.T0080.001) to alter long-term decisions.
AML.T0081	Modify AI Agent Configuration	Persistence	Altering local config files (e.g. openclaw.json, heartbeat.md) affecting agent pipelines, allowing persistent backdoors and safety-control bypasses.
AML.T0083	Credentials from Config	Credential Access	Harvesting API tokens, credentials, or secrets stored in predictable config directories of developer agents (e.g., ~/.claude/, ~/.cursor/, ~/.codex/).
AML.T0084	Discover AI Agent Config	Discovery	Probing local configuration directories and system paths to map active tool definitions, safety thresholds, and integrated capabilities.
AML.T0112.000	Machine Compromise: Local Agent	Impact	Adversary executing shell escapes and compromising host operating systems by tricking a local agent running with user principal permissions.
AML.T0096	AI Service API	Command & Control	Leveraging legitimate service and LLM orchestration APIs (e.g., OpenAI Assistants) as a covert, pre-authenticated control channel to bypass network detection.
AML.T0098	AI Agent Tool Cred Harvesting	Credential Access	Using the agent's broad tool integrations (OneDrive, SharePoint, local DBs) to harvest API keys and corporate secrets without human interaction.
AML.T0099	AI Agent Tool Data Poisoning	Initial Access	Placing malicious data or prompt-injection payloads in systems analyzed by agent tools (e.g., customer support tickets or repository READMEs) to hijack execution.
AML.T0100	AI Agent Clickbait	Initial Access	Exploiting how agentic browsers process HTML interfaces or visual markers, luring them to click malicious links, download files, or copy arbitrary code.
AML.T0101	Data Destruction via Tool Call	Impact	Abusing tool access to issue unlinks, file deletions, or database drops to cause denial of service (e.g., the April 2025 Claude Opus database deletion incident).
AML.T0086	Exfiltration via Tool Call	Exfiltration	Using legitimate outbound tool connections (web browsing, webhook APIs) to exfiltrate corporate credentials or model weights from the local machine.
AML.T0053	AI Agent Tool Invocation	Execution	Compromised agent spawning bash processes, executing shell commands, or abusing integrated tool capabilities to manipulate environment files.

Tech ID	ATLAS Technique Name	Strategic Tactic	Threat Vector & Technical Description
AML.T0103	Deploy AI Agent	Execution	Adversary deploying unauthorized AI agents onto target infrastructure to conduct automated, hands-free network discovery, zero-day research, and exfiltration.
AML.T0102	Generate Malicious Commands	Execution	Tricking developer-side AI coding copilots into generating and running dangerous system-level commands during CI/CD operations.

REAL-WORLD CASE STUDIES & RECOMMENDED DEFENSES

Grounded Incidents & OS-Layer Security Stack:

MITRE ATLAS documents 42 case studies mapping real attacks. These include **SesameOp (AML.CS0042)**, where malware repurposed legitimate LLM APIs for encrypted command and control, and **Slack AI (AML.CS0035)**, demonstrating how indirect prompt injection can lead to unauthorized credential harvesting. Local coding agent exploits are also mapped to **Cursor MCP (AML.CS0045)** and **ChatGPT Memory Poisoning (AML.CS0040)**. Recent 2026 security research demonstrates that because self-state modifications converge to identical VFS operations at the OS layer, conventional sandboxes fail to block them. The recommended defense is a **three-layer deterministic security stack**:

- **L3 Prevention:** Lock Instruction and Configuration files (SOUL.md, TOOLS.md, openclaw.json) as read-only at runtime.
- **B2 Anomaly Detection:** Apply workload-conditioned statistical detection over VFS feature tuples (op_type, size_delta, delta_t).
- **Continuous Recovery:** Run a 15-event backup interval snapshotting self-state files to ensure 100% remediation with minimal data loss.